MACDENNIS I. NWAEZE, M.Sc., GIT

Missouri City, TX 77489 | 713-235-0509 | nwaezemacdennis@yahoo.com | linkedin.com/in/macdennis-nwaeze

https://yardi-mac.github.io/Macdennis-Nwaeze-Professional-Portfolio/

GEOPHYSICS | DIGITAL GEOSCIENCE | SUBSURFACE DATA SCIENCE | FORMATION EVALUATION

PROFESSIONAL SUMMARY

Geophysicist and graduate researcher combining petroleum geophysics, wellsite geology, formation evaluation, geospatial sensing, and data science. Experienced with seismic interpretation and well-to-seismic ties, real-time drilling and petrophysical data, XRF/XRD, GNSS/GPS, Python, Linux, HPC, GIS, and cloud workflows. Proven field leadership and QA/QC experience with a focus on translating complex subsurface and sensor data into decision-ready technical products.

EDUCATION

University of Houston, Houston, TX –

M.S., Geo-Sensing & Systems Engineering, 2026-Present;
M.S., Geophysics (Petroleum Geophysics), 2022;
B.S., Geology, Minors in Mathematics & Geophysics, 2020

University of Texas at Austin - McCombs School of Business –

Post-Graduate Program, Cloud Computing (AWS & Microsoft Azure), 2025

TECHNICAL SKILLS

Geophysics & Subsurface: Seismic interpretation, well-to-seismic tie, AVO, well-log analysis, formation evaluation, mudlogging, petrophysics, reservoir characterization, XRF/XRD, Petrel, Petra, Oasis Montaj
Data & Computing: Python, MATLAB, SQL, Linux, Bash, HPC, GitHub, JSON, Excel, Power BI, AWS, Microsoft Azure
Geospatial & Sensing: GNSS/GPS, GNSS-IR, GipsyX, GIS/ArcGIS, remote sensing, LiDAR, ENVI, geospatial visualization
Certifications: ASBOG Geologist-in-Training (GIT), Rig Pass

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | National Center for Airborne Laser Mapping (NCALM), University of Houston | Houston, TX | Jan 2026-Present

- Process, QA/QC, and analyze large geospatial and engineering datasets using Python, Linux, HPC, GIS, and database workflows; build reproducible processing and visualization tools.
- Integrate GNSS, tide-gauge, satellite precipitation, atmospheric, and geospatial data for coastal-hazard research; prepare statistical analyses, technical figures, reports, and presentations.

QA/QC Geologist / Data Analyst | Diversified Well Logging LLC | Conroe, TX | Nov 2024-Jan 2026

- Analyzed real-time petrophysical and geochemical data using XRF/XRD; performed QA/QC to maintain reliable formation-evaluation and operational datasets.
- Developed technical reports, databases, and visualizations using Excel, SQL, and Power BI and integrated well-log, geochemical, and seismic information for interpretation.

Wellsite Geologist / Field Unit Manager | Diversified Well Logging LLC | South Texas | Feb 2023-Nov 2024

- Led field operations, safety, regulatory compliance, real-time geological/drilling data collection, gas and lithology interpretation, and daily reporting while coordinating with drilling and operations teams.
- Improved sample-collection workflow efficiency by approximately 33% using RoboLogger EZ automation; delivered mudlogs, formation-evaluation summaries, and wellsite technical reports.

SELECTED RESEARCH & TECHNICAL PROJECTS

NASA FINESST Proposal - Data-Driven Graph Forecasting of Hurricane-Induced Compound Flooding: Developing an attention-based spatiotemporal graph neural network and GIS framework integrating NASA GPM IMERG, SMAP, GNSS-derived IWV, tide gauges, river gages, and terrain/connectivity data to forecast Texas Gulf Coast flooding at 1-, 3-, 6-, and 12-hour lead times.

39th International Conference on Coastal Engineering (ICCE 2026), Galveston, TX — May 2026: Presented *"GNSS for Hurricane Monitoring: Tracking Storm Surge and Varying Atmosphere"*, integrating GNSS-IR sea-level retrievals, GNSS-derived atmospheric water vapor, NOAA tide gauges, and NASA GPM/IMERG precipitation to evaluate Hurricane Ian coastal and atmospheric response.

M.S. Geophysics Thesis - Delaware & Cooper Basins: Improved well-to-seismic tie accuracy through velocity-model corrections and AVO attribute analysis for reservoir characterization.

Mudlogging Copilot — AI-Powered Wellsite Geoscience Platform: Developing an AI-assisted wellsite platform integrating LAS/CSV drilling data, chromatograph gas, lithology, cuttings fluorescence, formation tops, and operational events. Uses digital image processing, CNN/MLP/transformer models to identify cuttings lithology and estimate lithologic proportions, while combining fluorescence and non-connection gas shows to identify depth-dependent hydrocarbon indicators during each drilling shift and provide decision support for mudloggers, operations geologists, company representatives, and reservoir engineers.

Geological Society of America Annual Meeting — GSA 2020 Connects Online: Coauthor, "Conversion of a Field Geodesy Module on the Neotectonics of the Northern Rocky Mountains into an Online Exercise," Paper 154-10. Applied GPS/geodesy, DEM and orthophoto analysis, ArcGIS fault mapping, tectonic geomorphology, and seismic-hazard analysis to active faults in the northern Rocky Mountains.

PROFESSIONAL AFFILIATIONS

AAPG (2019-Present) | ASCE (2025-Present) | NSBE Region 5 (2015-Present) | Geological Society of America (2019 – Present)